# Normal Distribution:

## Problems:

1. The marks of the students in a group follows normal distribution, with mean 96 and standard deviation 2.13. Compute the chance that the mark of a randomly selected student will lie between 95 and 99.

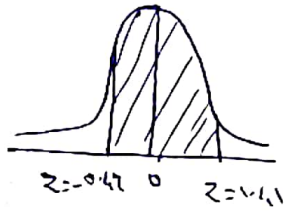
**Solution:**

Let $X$ represent marks.

Given $X \sim N(96, (2.13)^2)$ i.e., $\mu = 96$ and $\sigma = 2.13$

$$P(95 < X < 99) = P\left[\frac{95-\mu}{\sigma} < Z < \frac{99-\mu}{\sigma}\right]$$

$$= P\left[\frac{95-96}{2.13} < Z < \frac{99-96}{2.13}\right]$$




$$= P(-0.47 < Z < 1.41)$$

$$= P(-0.47 < Z < 0) + P(0 < Z < 1.41)$$

$$= P(0 < Z < 0.47) + P(0 < Z < 1.41)$$

$$= 0.1808 + 0.4207$$

$$= 0.6015.$$

2. 20% of the observations in the normal distribution are below 60. 80% of the observations are below 120. Compute the mean and standard deviation of the distribution.

**Solution:**

Let $X$ be the normal variate.

Given $P(X \le 60) = 20\%$.

i.e., $P\left(Z \le \frac{60-\mu}{\sigma}\right) = 0.2 \longrightarrow ①$

and $P(X \le 120) = 80\%$.

i.e., $P\left(Z \le \frac{120-\mu}{\sigma}\right) = 0.8 \longrightarrow ②$

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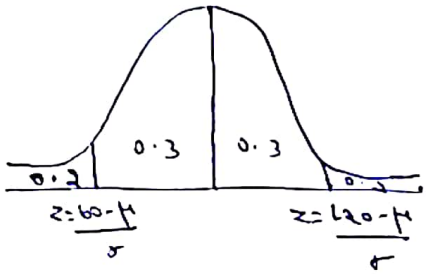
Representing ① and ② in the diagram




From the diagram

$$P\left(0 < z < \frac{120-\mu}{\sigma}\right) = 0.3$$

$$\Rightarrow \frac{120-\mu}{\sigma} = 0.85 \qquad [\because \text{from the tables}$$

$$\Rightarrow 120 - \mu = 0.85\sigma \qquad \longrightarrow ③$$

Again from the diagram

$$P\left(\frac{60-\mu}{\sigma} < z < 0\right) = 0.3$$

$$P\left(0 < z < \frac{\mu-60}{\sigma}\right) = 0.3 \qquad [\because P(-a<z<0) = P(0<z<a)$$

$$\Rightarrow \frac{\mu-60}{\sigma} = \cancel{0.3}\ 0.85 \quad [\because \text{from the tables}$$

$$\Rightarrow \mu - 60 = \cancel{0.3}\ 0.85\sigma. \longrightarrow ④.$$

Solving ③ and ④,

$$\mu = 90 \text{ and } \sigma = 39.29$$

3. If the actual amount of instant coffee which a filling machine puts into 6-ounce jars is a random variable having a normal distribution with SD = 0.05 ounce and if only 3% of the jars are to contain less than 6 ounces of coffee, what must be the mean fill of these jars?

Solution:

Let $x$ be the actual amount of coffee put into the jars.

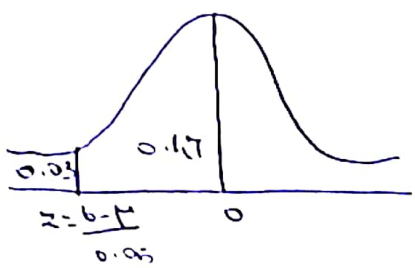
Then $x$ follows $N(\mu, 0.05)$ $\quad [\because \sigma = 0.05$

Given $P(x < b) = 3\%$.

$$P\left(z < \frac{b-\mu}{\sigma}\right) = 0.03$$

$$P\left(z < \frac{b-\mu}{0.05}\right) = 0.03$$

Represent this in the diagram




Now from the diagram we have.

$$P\left(\frac{b-\mu}{0.05} < z < 0\right) = 0.47$$

$$P\left(0 < z < \frac{\mu - b}{0.05}\right) = 0.47$$

$[\because P(-a < z < 0) = P(0 < z < a)$.

$$\Rightarrow \frac{\mu - b}{0.05} = 1.808 \quad [\because \text{From the tables.}$$

$$\mu - b = 0.0901$$

$$\mu = b + 0.0901 \text{ ounces} //.$$

4. The weekly wages of 1000 workmen are normally distributed around a mean of Rs.70 with a SD of Rs.5. Estimate the number of workers [whose weekly wages will be (i). between Rs.69 and Rs.72 (ii). less than Rs.69 (iii). more than Rs.72.] between Rs.70 and Rs.72.

$P(0 \le z \le 0.4) =$

Solution:

Let x be the RV denoting the weekly wages of a worker.

Given $\mu = 70$ and $\sigma = 5$.

The normal variate $z = \frac{x-\mu}{\sigma} = \frac{x-70}{5}$

(i). $P(69 < x < 72) = P\left(\frac{69-70}{5} < z < \frac{72-70}{5}\right)$

$= P(-0.2 < z < 0.4)$

$= P(-0.2 < z < 0) + P(0 < z < 0.4)$

$= P(0 < z < 0.2) + P(0 < z < 0.4)$

$= 0.0793 + 0.1554$

$= 0.2347$

Out of 1000 workmen, the number of workers whose wages lies between Rs.69 and Rs.72

$= 1000 \times P(69 < x < 72)$

$= 1000 \times 0.2347$

$= 234.7$

$\simeq 235.$

(ii). P (less than Rs.69) $= P(x < 69)$

$= P\left(z < \frac{69-\mu}{\sigma}\right)$

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$$= P\left(z < \frac{69-70}{5}\right)$$

$$= P(z < -0.2)$$

$$= 0.5 - P(0 < z < 0.2)$$

$$= 0.5 - 0.0793$$

$$= 0.4207.$$

Out of 1000 workmen, the number of workers whose wages earners than Rs. 69

$$= 1000 \times P(X < 69)$$

$$= 1000 \times 0.4207$$

$$= 420.7$$

$$\simeq 421.$$

(iii), $P(\text{more than Rs. } 72) = P(X > 72)$

$$= P\left(z > \frac{72-\mu}{\sigma}\right)$$

$$= P\left(z > \frac{72-70}{5}\right)$$

$$= P(z > 0.4)$$

$$= 0.5 - P(0 < z < 0.4)$$

$$= 0.5 - 0.1554$$

$$= 0.3446$$

Out of 1000 workmen, the number of workers whose wages are more than 72

$$= 1000 \times P(X > 72)$$

$$= 1000 \times 0.3446$$

$$= 344.6$$

$$= 345.$$

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5. The average sales of a commodity in a particular locality is known to be 46 kg with a sd of 2 kg. Assuming normal distribution, find the probability that (i) the sales will be between 40 kg and 42 kg (ii) the sales will be greater than 44 kg.

Solution:

Let $x$ be the RV describing the average sales.

Given $\mu = 46$ and $\sigma = 2$.

The standard normal variate $z = \frac{x-\mu}{\sigma} = \frac{x-46}{2}$

(i). $P(40 < x < 42) = P\left(\frac{40-46}{2} < z < \frac{42-46}{2}\right)$

$= P(-3 < z < -2)$

$= P(2 > z > 3)$

$= P(0 < z < 3) - P(0 < z < 2)$

$= 0.4987 - 0.4772$

$= 0.0215$.

(ii). $P(x > 44) = P\left(z > \frac{44-46}{2}\right)$

$= P(z > -1)$

$= 0.5 + P(0 < z < 1)$

$= 0.5 + 0.3413$

$= 0.8413$.

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